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The Mesoproterozoic Zig-Zag Dal basalts and associated intrusions of eastern North Greenland: mantle plume–lithosphere interaction

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Abstract The lavas of the Zig-Zag Dal Formation of eastern North Greenland constitute a Mesoproterozoic tholeiitic flood basalt succession up to 1,350 m thick, extending >10,000 km^2, and underlain by a sill complex. U–Pb dating on baddeleyite from one of the sills thought to be contemporaneous with the lava extrusion, gives an age of 1,382 ± 2 Ma. The lavas, subdivided from oldest to youngest into Basal, Aphyric and Porphyritic units, are dominantly basaltic (>6 wt.% MgO), with more evolved lavas occurring within the Aphyric unit. The most magnesian lavas occur in the Basal unit and the basaltic lavas exhibit a generalised upward decrease in Mg number (MgO/(MgO + $Fe_2O_3^T$)) through the succession. All of the lavas are regarded as products of variable degrees of olivine, augite and plagioclase fractionation and to be residual after generation of cumulates in the deep crust. The basaltic lavas display an up-section fall in the ratio of light to heavy rare-earth elements (LREE/HREE) but an up-section rise in Zr/Nb, Sc, Y and HREE. The older lavas (Basal and Aphyric units) are characterised by low ε_{Nd} and ε_{Hf} in contrast to higher values in the younger (Porphyritic unit) lavas. The Porphyritic Unit basalts are characterised by a notable enrichment in Fe and Ti. The Zig-Zag Dal succession is inferred to reflect an increase in melt fraction in the sub-lithospheric mantle, with melting commencing in garnet–lherzolite facies peridotites and subsequently involving spinel-facies mantle at increasingly shallow depths. Melting is deduced to have occurred beneath an attenuating continental lithosphere in conjunction with ascent of a mantle plume. Lithospheric contamination of primitive melts is inferred to have diminished with time with the Porphyritic unit basalts being products of essentially uncontaminated plume-source magmas. The high iron signature may reflect a relatively iron-rich plume source.

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Introduction

In northern East Greenland between 80°N and 83°N, a thick (>8,000 m) well-preserved sequence of Precambrian and lower Palaeozoic strata is exposed which has experienced only gentle tilting and open folding since its formation (Henriksen 1979, 1980). This sequence is bounded to the south-east by the westward directed thrusts that define the north-western margin of the Caledonide orogenic belt. Eastern North Greenland is thus divisible into a relatively undeformed foreland sequence to the north-west of the Danmark Fjord and a highly deformed orogenic belt to the south-east.

Within the foreland sequence (Fig. 1) the Zig-Zag Dal lavas are a remnant of a more extensive flood basalt sequence whose south-western extension has been lost by erosion and whose north-eastern extension is largely covered by younger sedimentary formations. Although earlier work on the petrology of the sequence was presented by Kalsbeek and Jepsen (1984) it was considered that, in the light of current interest in large igneous

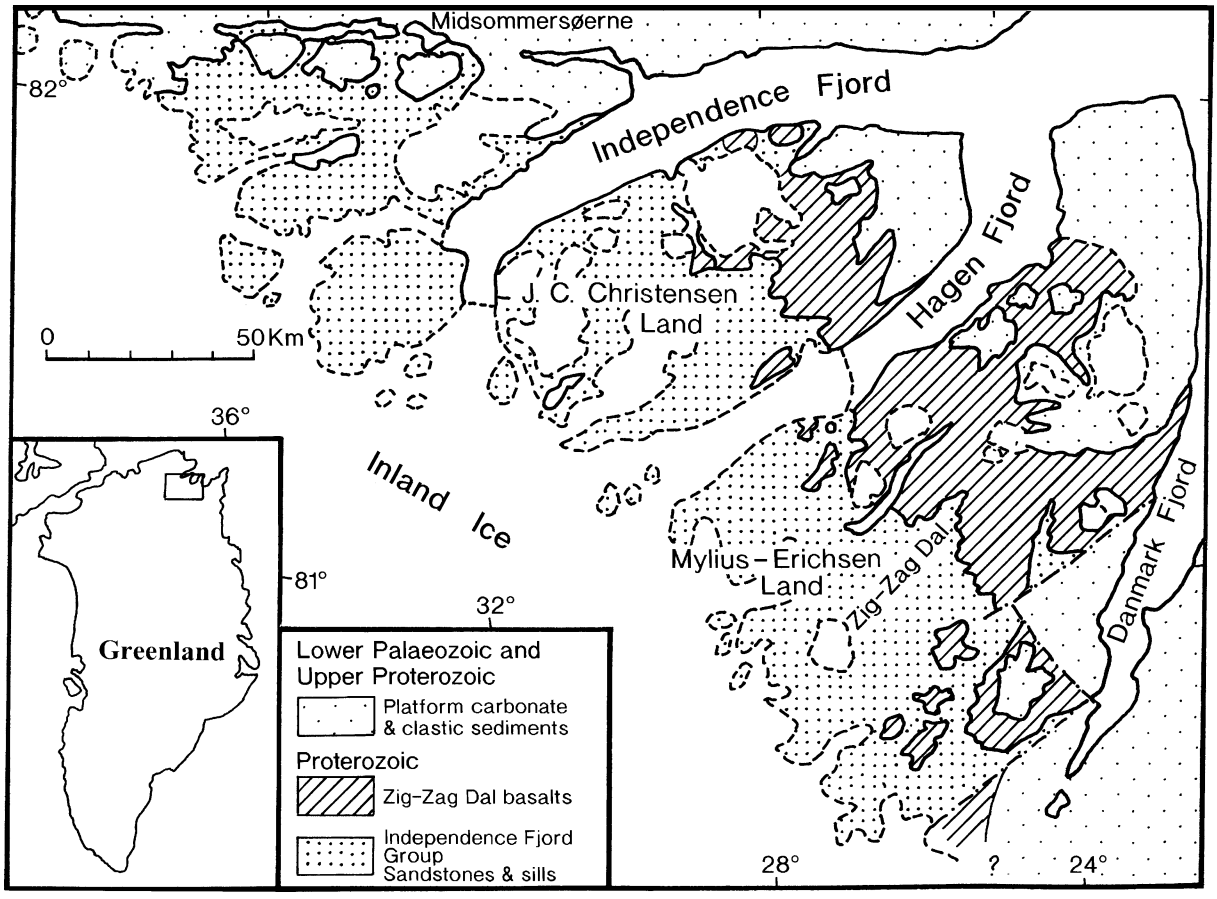


Fig. 1 Geological sketch map showing outcrop areas of Zig-Zag Dal lavas and Independence Fjord Formation. (After Kalsbeek and Jepsen 1984)

provinces and evidence for Precambrian hot-spot activity, re-examination of this remote and very inaccessible sequence was merited.

The new investigation suggests progressive melting of an evolving mantle plume. Marked iron-enrichment of the upper part of the succession is taken to signify participation of more iron-rich mantle sources. The data accord with the belief that the underlying Midsommersø sills were partly coeval with the lavas and a new precise U–Pb (baddeleyite) date for a sample of the sill dolerites (1,382 ± 2 Ma) is adopted as the best estimate for the age of the Zig-Zag Dal lavas.



Fig. 2 The type-section of the lavas in Zig-Zag Dal

The Zig-Zag Dal basalts

The Zig-Zag Dal basalts were described as a mid-Proterozoic lava sequence cropping out over some 10,000 km^2 and representing a classic tholeiitic continental flood basalt sequence (Kalsbeek and Jepsen 1984). The type-section is superbly exposed in the deep glaciated valley called Zig-Zag Dal (Figs. 1, 2). It is predominantly made up of subaerial lavas which conformably overlie a sequence of sedimentary rocks over 2 km thick (the Independence Fjord Formation; Collinson 1980, 1983). This Formation comprises cross-bedded, arkosic and abundantly ripple-marked sandstones of deltaic facies, interpreted to overlie a 'crystalline basement' which, however, is not exposed in the region. Although the maximum thickness recorded by Kalsbeek and Jepsen (1984) for the basaltic sequence is 1350 m, the lavas experienced variable degrees of erosion prior to the deposition of the overlying late Proterozoic Campanuladal Formation sandstones. The Zig-Zag Dal basalts were inferred by Kalsbeek and Jepsen (1984) to occupy a basin, peneplaned at the top,

whose central area underwent subsidence after or accompanying the volcanism. The thicker lavas of the Zig-Zag Dal succession have columnar jointing with well developed colonnade and entablature structures and the inference is that the entire sequence was erupted in a down-sinking basin and subject to varying degrees of water-cooling. These observations accord with the concept of a continental terrane that was subjected to extension and lithospheric thinning.

A profusion of dolerite intrusions (conformable sills, dykes and inclined sheets known as the Midsommersø dolerites), that may have been contemporaneous and cogenetic with the lavas, is found within the underlying Independence Fjord Formation. Whilst the Midsommersø dolerites gave well-defined Rb–Sr whole-rock isochron ages of ca. 1,230 Ma for some of the intrusions (Kalsbeek and Jepsen 1983), the U–Pb baddeleyite date obtained in the present study indicates an older emplacement age of 1,382 Ma (see below). Palaeomagnetic data support contemporaneity of the dolerites and the Zig-Zag Dal basalts (Marcussen and Abrahamsen 1983).

The Zig-Zag Dal lavas were subdivided by Jepsen et al. (1980) into three units; (1) Basal, (2) Aphyric and (3) Porphyritic. The Basal unit (100–120 m thick) includes pillow lavas and was followed by deposition of thin sandstones and dolomites. The overlying Aphyric unit (390–440 m thick) and the stratigraphically youngest lavas (Porphyritic unit, up to 750 m thick) comprise approximately 30 lavas with thicknesses ranging from 10 to 120 m. The lower part of the Porphyritic unit contains thin, laterally impersistent lavas and intercalations of conglomerate, sandstone and dolomite. Tuff horizons and localised pillow basalts also occur in the lowermost parts of the Porphyritic unit. Within both the Aphyric and Porphyritic units there are lavas that can be traced laterally along tens of kilometres of outcrop. Particularly notable among these in the Aphyric unit are (a) a distinctive lava No. 52, (the "brown marker") at ca. 300 m above the base and (b) the basaltic lava beneath it (No. 51) which was estimated to have a minimum volume of 600 km^3 (Kalsbeek and Jepsen 1984). Most of the lavas are basaltic although the 'brown marker' flow is a basaltic andesite (54.5 wt.% SiO_2 and 4.42 wt.% MgO). Furthermore, elsewhere in northern J. C. Christensen Land and SE Mylius-Erichsen Land, Kalsbeek and Jepsen (1984) noted still more evolved compositions (rhyolites) within the Aphyric unit.

Deeply weathered lava tops and palaeosols are absent or poorly developed in the Zig-Zag Dal lavas and vestiges of corded pahoehoe textures remain on some lava surfaces. These field observations suggest that the entire lava pile accumulated within a short time (Kalsbeek and Jepsen 1984); palaeomagnetic data (Marcussen and Abrahamsen 1983) support this interpretation. Lavas of the Basal unit have micro-phenocrysts of plagioclase, augite and (pseudomorphed) olivine. In the Aphyric unit, there are micro-phenocrysts of augite and plagioclase. The Porphyritic unit is characterised by larger euhedral phenocrysts of plagioclase and porphyritic aggregates of plagioclase and augite. No dykes were observed within the Zig-Zag Dal succession and the probability is that the exposed lavas are remote from their eruptive vents.

Amygdales in many of the lavas contain quartz and/or epidote linings enclosing chlorite, with later calcite. Additionally, botryoidal prehnite occurs, frequently intergrown with native copper and secondary copper carbonates. The lavas have been variably affected by low temperature hydrothermal metamorphism resulting in development of calc-silicate and mafic phyllosilicate-bearing assemblages (Bevins et al. 1991). This zeolite facies to lowermost prehnite-pumpellyite facies metamorphism indicates temperatures of up to 200°C, and is inferred to have taken place simultaneous with, or shortly after cessation of, volcanism.

The intermittent occurrence of pillow lavas with intercalated sediments and tuffs implies that the depositional environment varied between shallow subaqueous and subaerial on a steadily subsiding, near-horizontal platform. Episodic interruptions in the magma supply gave quiescent interludes during which sedimentation resumed. A broad delta region is envisaged within which subsidence and clastic sedimentation took place over a long period prior to the onset of magmatism. There is no evidence for any uplift immediately prior to the volcanism.

The Midsommersø dolerites

The sandstones beneath the lava succession are intruded by large numbers of sills whose estimated total thickness locally exceeds 1,000 m (Kalsbeek and Jepsen 1983). Consequently the combined thickness of Zig-Zag Dal lavas and underlying intrusions can exceed 2 km. These intrusions, predominantly of doleritic composition, are referred to as the Midsommersø sills. Kalsbeek and Jepsen (1983) subdivided them into three categories; *viz*. (1) (tholeiitic) dolerites; (2) granophyric dolerites (with gradation from dolerite to granophyre); (3) highly siliceous mobilised sandstones (rheopsammites). Wide compositional variation in the intrusions was ascribed to varying degrees of fractional crystallisation, crustal melting and assimilation processes. Kalsbeek and Jepsen (1983) described large-scale generation of silicic anatectic melts in association with the Midsommersø dolerites. As with the extrusive succession, many of the dolerites also experienced intensive hydrothermal alteration deduced to have been associated with the magmatism. This alteration has caused significant chemical changes and reddening of some of the dolerites (Kalsbeek and Jepsen 1983).

Age of the lavas and sills

Although attempts to date the lavas have been fruitless, Rb–Sr whole-rock isochrons of the Midsommersø

dolerites gave an age of ~1,230 Ma (Kalsbeek and Jepsen 1983). An age of 1,230 ± 20 Ma was obtained for a metasomatically altered 'red dolerite' and an age of 1,220 ± 25 Ma for a rheopsammitic intrusion (Kalsbeek and Jepsen 1983). Recent re-analysis of the samples of 'red dolerite' (both $^{87}Rb/^{86}Sr$ and $^{87}Sr/^{86}Sr$) gave the same result and the 1,230 Ma date is regarded as reliable, although its interpretation is not straightforward. Since the rocks have been affected by hydrothermal activity resulting in complete Sr-isotope homogenisation and the hydrothermal processes are likely to have been related to the magmatism, the isochron was interpreted indirectly to date the age of intrusion. Kalsbeek and Jepsen (1984) assumed contemporaneity for the intrusive and extrusive suites and the similarity of palaeomagnetic pole positions of the two suites lent support to this supposition (Marcussen and Abrahamsen 1983). Rb–Sr isotopic studies on clay minerals from a siltstone within the Independence Fjord Group (southern J. C. Christensen Land; Larsen and Graff-Petersen 1980) gave a minimum depositional age of ca. 1,380 Ma although the authors could not quantify the reliability of the age.

In order to obtain a more precise age for the Midsommersø dolerites, a U–Pb analysis was carried out on separated baddeleyite. A suitable (fresh and uncontaminated) dolerite sample (GGU 273242) was pulverised using standard crushing equipment (jaw crusher, disk mill) and a heavy mineral concentrate was recovered from this powder using a Wilfley table. Baddeleyite was concentrated using a series of mineral separation steps that included magnetic separation (Frantz Isodynamic Separator) and density (methylene iodide) techniques. Approximately 300 euhedral, tan to brown baddeleyite crystals were recovered, most of which are slender blades with lengths of 20–40 μm. The procedures for purifying uranium and lead from baddeleyite and their isotopic ratio measurement using a VG354 thermal ionisation mass spectrometer are outlined elsewhere (Heaman and Machado 1992; Heaman et al. 2002).

The U–Pb results for three small multi-grain baddeleyite fractions are reported in Table 1 and on a concordia diagram in Fig. 3. Baddeleyite fraction #1 consisted of eleven larger, dark brown, anhedral crystals; fraction #2 consisted of one hundred and fifty-three smaller tan blades, some of which contained mineral inclusions and fractures, and fraction #3 consisted of 67

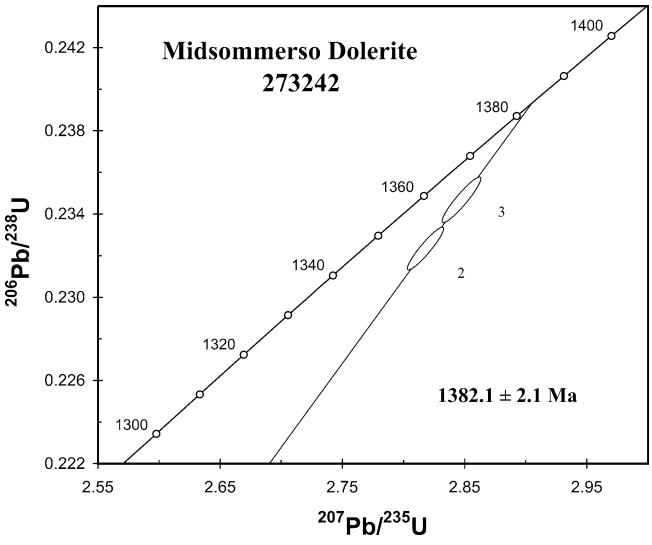


Fig. 3 Concordia plot of $^{206}Pb/^{238}U$ versus $^{207}Pb/^{235}U$ for baddeleyite from Midsommersø dolerite sample 273242

small, light brown blades. Baddeleyite in this sample generally has lower uranium contents (192–297 ppm) than most baddeleyites from mafic rocks but the Th/U value is very constant at ~0.1 which is typical for such samples. The three baddeleyite analyses are slightly discordant (0.4, 1.9 and 2.8%, respectively) but yield very similar $^{207}Pb/^{206}Pb$ dates of 1381.9, 1382.4 and 1382.1 Ma. The weighted mean $^{207}Pb/^{206}Pb$ date of 1382.1 ± 1.8 Ma is identical to the upper intercept date of 1382.1 ± 2.6 Ma determined by a linear regression treatment of all three analyses.

In view of the uncertainty in the interpretation of the Rb–Sr isochron data we provisionally adopt the 1,382 Ma baddeleyite age as the best estimate of the age of the Zig-Zag Dal basalts. However, in view of the discrepancy between this age and the Rb–Sr date of ~1,230 Ma, the possibility remains that the sill complex involves intrusions of differing ages.

Analytical methods

The samples studied were collected during field-work by the Geological Survey of Greenland (GGU; now incorporated into the Geological Survey of Denmark and

Table 1 U-Pb baddeleyite results for the midsommerso gabbro sill, Greenland

	Weight (μg)	U (ppm)	Th (ppm)	Pb (ppm)	Th/U	TCPb (pg)	$^{206}Pb/^{204}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	$^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	$^{207}Pb/^{206}Pb$	Disc (%)
273242														
2 B 8		297	31	67	0.10	10	3,772	0.23236 ± 42	2.8185 ± 58	0.08797 ± 6	1346.9 ± 2.2	1,360.5 ± 1.5	1,381.9 ± 1.4	2.8
3 B 51		228	23	53	0.10	100	1,714	0.23468 ± 42	2.8472 ± 60	0.08799 ± 8	1359.0 ± 2.2	1,368.1 ± 1.6	1,382.4 ± 1.7	1.9

Notes: Mineral Analysed: B baddeleyite,Th concentration estimated from abundance of ^{208}Pb and corresponding $^{207}Pb/^{206}Pb$ ages, TCPb refers to the total amount of common Pb in picograms measured in the analysis, Atomic ratios are corrected for fractionation (0.1%/amu), blank (2 pg Pb; 0.5 pg U), spike and common Pb (depleted mantle), $^{206}Pb/^{204}Pb$ ratios are corrected for fractionation and spike onlyAll errors quoted in this table are 1 sigma uncertainties

Greenland, GEUS) in 1978, and the sample numbers refer to the GGU sample files. New whole-rock analyses were carried out by XRF at the University of Edinburgh, using a Philips PW 1480 spectrometer equipped with a Rh-anode X-ray tube. The instrument was calibrated using CRPG and USGS reference standards (Govindaraju 1994). Major elements were determined on fused glass discs with corrections applied for inter-element mass absorption effects. Trace elements were determined on pressed powder discs and count rates were corrected for line-overlap and mass absorption effects. A full description of the techniques used, including analytical conditions and precision estimates, is given in Fitton et al. (1998). Twenty-one samples were also analysed for REE by ICP-MS spectrometry at Royal Holloway, University of London.

The Nd isotope analyses on 14 samples were performed at the unit for Isotope Geology, Geological Survey of Finland by OTR. Rock powders (200–250 mg) were spiked with a ^{149}Sm–^{150}Nd tracer and dissolved in moulded Teflon vials in a 1:4 mixture of HNO_3 and HF for a minimum of 48 h. After evaporation, the samples were dissolved in HCl. Light rare earth elements (REE) were separated using standard cation exchange chromatography and Sm and Nd were purified on quartz columns according to the method of Richard et al. (1976). The total procedural blank was <300 pg for Nd. Isotope ratios of Sm and Nd were measured on a VG Sector 54 mass spectrometer (those of Nd in dynamic mode). Repeated analyses of the La Jolla Nd standard gave $^{143}Nd/^{144}Nd$ of 0.511850 ± 0.000015 (mean and external 2 sigma error of 10 measurements); the external error in the reported ^{143}Nd–^{144}Nd ratios is estimated to be better than 0.0025%.

For the Lu–Hf isotope analyses, samples were dissolved in steel-jacketed Teflon bombs with a mixed ^{176}Lu–^{180}Hf spike added to the sample powders at the outset of the dissolution procedure. Upon complete sample-spike homogenisation, a Hf-bearing fraction was separated from a REE-bearing fraction on a cation-exchange column. A Yb–Lu fraction was isolated from the latter on an HDEHP column, while Hf was further purified, first through an anion-exchange column to remove remaining matrix elements, then through a cation-exchange column separating Ti and some Zr from the Hf (Blichert-Toft et al. 1997; Blichert-Toft 2001). Total procedural Lu and Hf blanks were <20 and <25 pg, respectively.

The isotopic analyses of Hf and Lu were carried out by MC-ICP-MS on the VG Plasma 54 instrument in Lyon following the procedures described in Blichert-Toft et al. (1997) and Blichert-Toft (2001). In order to monitor machine performance, the JMC-475 Hf standard was run systematically before and after each sample and gave, throughout this study, 0.282160 ± 0.000010 for $^{176}Hf/^{177}Hf$ (two standard deviations), corresponding to an external reproducibility of 35 ppm.

Geochemistry

Seventy samples from the Zig-Zag Dal Formation and 37 from the Midsommersø intrusions were analysed by XRF spectrometry. These include 57 samples from the Zig-Zag Dal lavas which were previously analysed and reported by Kalsbeek and Jepsen (1984). The analytical data thus cover a broader range than the original study. Representative whole-rock analyses are provided in Table 2.

No total alkali silica diagram is presented in view of the pervasive alteration of the lavas and the consequent unreliability of the sodium and potassium contents. There is, however, no reason to doubt the overall tholeiitic character of the lavas and intrusions; pigeonite is occasionally preserved in some of the fresher samples. The lavas in the 1,350 m type-section are all basaltic except for the distinctly more evolved tholeiitic andesite 'brown marker' in the Aphyric unit. The following discussion will be restricted to the basalts (i.e. the lavas with $MgO > 6$ wt.%) in the type-section where the stratigraphic relationships are known with confidence.

Although the concentrations of all elements are, to some extent, influenced by the mobility of Na, K, Ca and volatiles (Kalsbeek and Jepsen 1984), this effect is most pronounced in the Basal unit where the proportion of volatiles is highest (3.4–7.8 wt.%). The concentrations of the more immobile elements in the Aphyric and Porphyritic units probably more closely reflect original magmatic compositions. As described below, many of the elements show distinct trends through the lava sequence, superimposed on a within-unit scatter that can, at least partly, be interpreted as the result of variable degrees of pre-eruption crystal fractionation.

There is a very generalised up-sequence decrease in MgO content from the Basal unit comprising basalts with 9.0–10.7 wt.% MgO (on volatile-free basis), through more erratic values in the Aphyric unit to reasonably constant values between 6.6 and 8.3 wt.% in the Porphyritic unit. A plot of Mg number ($100\ MgO/(MgO + Fe_2O_3^T)$ versus height (Fig. 4a) shows an overall decline through the sequence with a less erratic pattern than that for MgO. Mg numbers decrease sharply from the Basal unit (54–46) to the Aphyric unit, but stay constant in the Aphyric unit at a value of 46 until approximately 500 m up-sequence. At this horizon values fall to around 40 from the upper part of the Aphyric unit to the top of the succession.

There is also a tendency for silica to decrease slightly from 52 to 55 wt.% in the Basal unit, through highly variable contents (50–54%) in the Aphyric unit, to 49–51% in the Porphyritic unit. Total iron ($Fe_2O_3^T$) increases up-section (Fig. 4a). The $Fe_2O_3^T$ rises within the Basal unit, but remains roughly constant through the Aphyric unit lavas until, above the 'brown marker' at ~500 m in the sequence, it rises in the higher parts of the Aphyric unit attaining its highest values (up to 14.8 wt.%) in the Porphyritic unit. Although the stratigraphic behaviour of TiO_2 is somewhat more variable, it

Table 2 Representative analyses of basalts and dolerites from the Zig-Zag Dal lavas and Midsommersø sills

	Basal Series			Aphyric Series								Porphyritic Series								Midsommersø dolerites			
	273405	273407	273410	273419	273420	273426	273427	273430	273434	273436	273439	273442	273446	273447	273451	273470	273473	273478	273479	197402	335774	273251	273493
wt.%																							
SiO_2	49.78	51.49	50.77	50.65	51.04	54.57	54.39	50.07	48.82	50.87	50.39	50.03	48.85	47.89	47.60	49.54	49.25	48.88	48.76	51.11	47.57	49.62	60.59
Al_2O_3	14.49	14.96	14.49	13.96	14.05	13.33	13.06	14.48	14.77	13.87	14.28	14.07	14.09	13.95	13.47	13.73	14.24	14.72	14.52	13.95	13.84	11.06	12.95
Fe_2O_3*	10.22	9.55	10.45	10.67	10.61	12.24	11.84	10.61	10.54	11.68	11.19	12.96	12.66	12.40	12.45	13.37	13.11	12.64	12.70	11.32	14.74	20.33	7.14
MgO	10.19	9.44	8.93	8.19	8.06	4.69	4.16	8.78	8.37	6.84	7.41	6.69	7.48	7.84	6.63	7.13	7.32	7.51	8.18	8.30	5.87	3.67	4.57
CaO	6.25	2.76	5.41	8.55	8.50	5.50	6.36	12.43	10.98	7.83	9.91	11.19	9.13	8.05	11.11	11.57	11.24	12.02	10.98	8.94	9.95	8.52	5.27
Na_2O	2.62	4.62	3.99	2.76	3.26	3.48	3.60	1.68	2.92	4.04	2.78	2.20	3.32	3.73	1.97	2.03	2.23	1.88	1.89	1.76	2.23	2.51	3.61
K_2O	0.71	0.87	1.41	1.75	1.29	2.43	2.40	0.10	0.18	0.93	0.67	0.41	0.45	0.19	0.10	0.26	0.31	0.10	0.37	1.07	1.17	0.55	2.62
TiO_2	0.82	0.83	0.80	0.89	0.90	1.27	1.22	0.80	0.81	1.04	0.99	1.19	1.00	0.94	1.00	1.04	1.06	1.00	0.99	0.93	2.07	2.38	0.57
MnO	0.16	0.17	0.22	0.18	0.16	0.20	0.17	0.16	0.19	0.22	0.17	0.19	0.23	0.21	0.19	0.19	0.19	0.20	0.23	0.19	0.22	0.29	0.11
P_2O_5	0.13	0.13	0.12	0.14	0.14	0.19	0.18	0.09	0.09	0.13	0.12	0.13	0.09	0.09	0.10	0.10	0.10	0.09	0.09	0.10	0.26	0.23	0.09
LOI	4.51	4.76	3.40	1.97	2.11	1.94	2.15	1.14	2.51	2.43	2.12	0.63	2.51	2.89	4.13	0.84	0.83	0.96	0.78	2.04	1.52	0.67	2.14
Total	99.86	99.59	99.99	99.7	100.12	99.83	99.54	100.33	100.18	99.87	100.04	99.67	99.81	98.18	98.75	99.8	99.88	100	99.49	99.70	99.45	99.83	99.65
ppm																							
Nb	6.5	7.0	7.1	7.8	7.3	19.1	18.1	4.1	3.7	7.0	4.9	5.5	2.7	2.3	3.5	3.8	4.2	3.0	2.9	4.7	14.7	7.9	6.8
Zr	76.9	84.6	83.3	117.2	116.5	227.2	223.1	65.3	66.5	106.2	83.3	87.6	59.4	58.9	73.6	70.3	69.8	65.1	65.2	88.1	158.4	157.0	103.3
Y	15.4	15.7	16.9	23.7	24.1	39.7	38.7	18.5	17.6	23.7	19.7	24.3	20.4	20.7	24.3	23.0	23.5	22.1	21.9	24.6	33.5	51.1	17.8
Sr	208.1	106.0	240.9	241.4	262.4	160.8	158.2	163.0	610.5	326.2	253.0	200.1	380.3	456.8	176.6	117.5	121.5	122.5	118.1	148.8	271.5	169.8	148.8
Rb	13.7	20.5	30.0	59.6	36.7	62.9	61.3	0.3	4.0	17.9	17.5	8.1	7.4	2.9	0.7	5.3	5.1	2.8	7.2	29.7	30.8	12.4	67.2
Th	1.2	0.6	1.1	2.1	2.4	6.5	6.5	1.4	1.1	2.1	1.4	0.7	0.8	0.5	0.6	1.1	0.6	1.1	0.7	1.6	2.9	1.9	3.6
Pb	3.7	2.2	2.5	3.6	3.2	5.6	6.5	0.5	11.0	9.6	3.3	1.7	2.9	2.9	0.8	1.2	2.0	1.1	1.9	5.2	2.9	3.9	4.0
Zn	75.0	80.1	85.8	77.9	80.7	103.9	31.9	72.7	77.3	102.1	78.5	91.6	82.3	80.7	91.1	83.7	92.1	85.0	92.9	69.1	125.0	166.6	13.7
Cu	92.1	87.5	75.1	96.5	94.9	127.3	73.8	105.7	47.0	94.9	55.2	115.5	156.7	147.2	157.7	105.3	156.5	163.9	522.6	124.2	177.1	226.9	32.4
Ni	85.8	89.9	102.1	135.7	135.7	30.0	30.6	127.0	131.2	75.0	112.0	94.4	138.8	147.1	97.5	102.3	131.9	144.4	143.2	105.2	90.6	11.8	62.4
Cr	339.1	383.5	399.6	354.2	355.7	4.2	9.4	353.1	346.5	147.9	223.8	106.2	306.2	320.9	244.7	265.7	256.8	271.2	277.8	111.2	158.4	5.9	62.4
V	240.5	247.0	230.4	260.1	258.2	326.2	316.7	279.3	280.1	311.6	290.9	354.4	365.5	333.0	366.2	382.6	358.4	360.8	359.6	311.5	416.2	467.1	86.5
Ba	230.8	77.2	517.4	558.6	444.5	582.5	461.5	69.3	219.4	309.2	219.5	163.6	437.7	142.9	91.5	78.9	68.4	34.2	77.9	242.8	436.4	214.0	152.0
Sc	36.9	40.2	35.1	39.9	40.5	39.2	38.1	39.7	41.9	41.0	40.7	44.1	43.5	39.8	44.6	47.8	45.4	44.3	47.3	47.1	41.7	49.8	78.9
La	10.3	11.5	11.3	15.6	15.0	35.8	32.7	8.3	7.6	ND	10.8	ND	5.6	5.7	7.5	ND	5.8	5.3	5.0	10.3	19.5	14.7	ND
Ce	22.4	24.9	24.1	33.1	32.3	74.0	70.6	18.4	17.5	ND	24.0	ND	14.1	13.5	18.4	ND	13.7	13.7	12.7	23.4	44.6	34.7	ND
Pr	2.7	3.2	3.2	3.7	3.7	8.0	8.0	2.2	2.1	ND	2.8	ND	2.0	1.8	2.2	ND	1.9	1.9	2.0	2.8	5.6	4.5	ND
ND	12.1	14.5	14.2	16.7	17.1	34.5	32.0	10.5	10.2	ND	14.1	ND	9.5	9.2	11.5	ND	10.5	9.6	10.3	12.7	25.1	22.5	ND
Sm	2.5	2.8	2.8	3.6	3.6	6.6	6.2	2.4	2.4	ND	3.1	ND	2.3	2.3	2.8	ND	2.6	2.5	2.7	3.0	5.5	5.8	ND
Eu	1.0	1.1	1.1	1.1	1.1	1.8	1.6	0.9	0.8	ND	1.1	ND	1.0	0.9	1.1	ND	1.0	1.0	1.0	1.0	1.8	2.1	ND
Gd	2.7	2.7	2.7	3.8	3.7	6.2	6.5	3.0	2.9	ND	3.3	ND	3.2	3.2	3.6	ND	3.5	3.5	3.2	3.7	6.1	7.6	ND
Dy	2.7	2.8	2.8	4.0	3.9	6.7	6.6	3.0	3.2	ND	3.5	ND	3.6	3.5	4.0	ND	4.0	3.8	3.8	4.3	6.0	8.5	ND
Ho	0.6	0.6	0.6	0.9	0.8	1.4	1.4	0.7	0.7	ND	0.7	ND	0.8	0.8	0.9	ND	0.8	0.8	0.8	0.9	1.2	1.8	ND
Er	1.6	1.8	1.8	2.7	2.5	4.1	3.7	2.0	2.0	ND	2.2	ND	2.3	2.0	2.6	ND	2.6	2.4	2.5	2.5	3.3	4.9	ND
Yb	1.5	1.6	1.6	2.4	2.3	4.0	3.7	1.8	1.9	ND	2.0	ND	2.3	2.1	2.4	ND	2.3	2.2	2.2	2.4	3.1	4.7	ND
Lu	0.2	0.3	0.3	0.4	0.4	0.6	0.6	0.3	0.3	ND	0.3	ND	0.4	0.3	0.4	ND	0.4	0.4	0.3	0.4	0.5	0.8	ND
height (m.)	5	13.75	35	141.25	165	310	318.75	376	453	486.25	546.25	580	735	785	853.8	1181.3	1202.5	1272.5	1302.5				

*Total Fe as Fe_2O_3
ND not determined

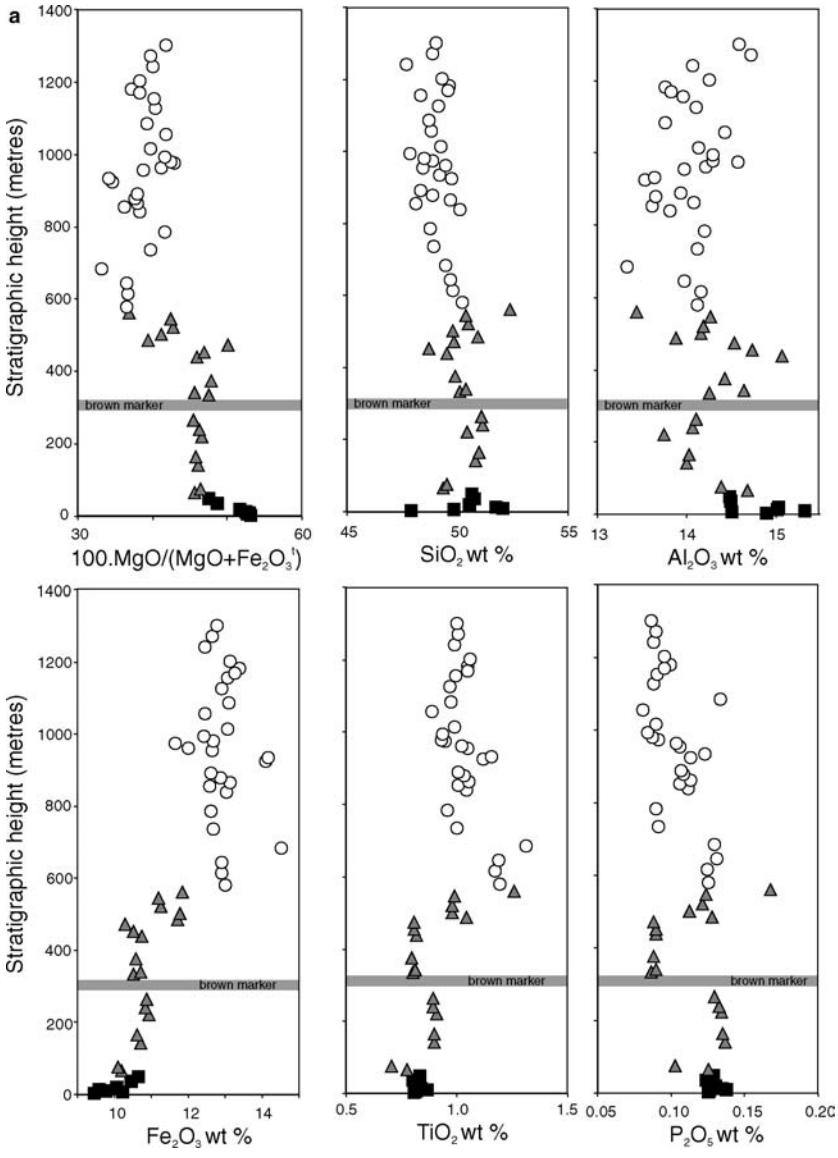


Fig. 4 Plots of major and trace element data versus stratigraphic numbers plotted against stratigraphic height in the type section; **a** Mg number, SiO_2, Al_2O_3, $Fe_2O_3^T$, TiO_2 and P_2O5; **b** Zr, Nb, V, Y, Sc and Cr. Basal unit, *solid squares*; Aphyric unit *grey triangles*; **c** Porphyritic unit, *open circles*. The height of the 'brown marker' is indicated by a *grey band*

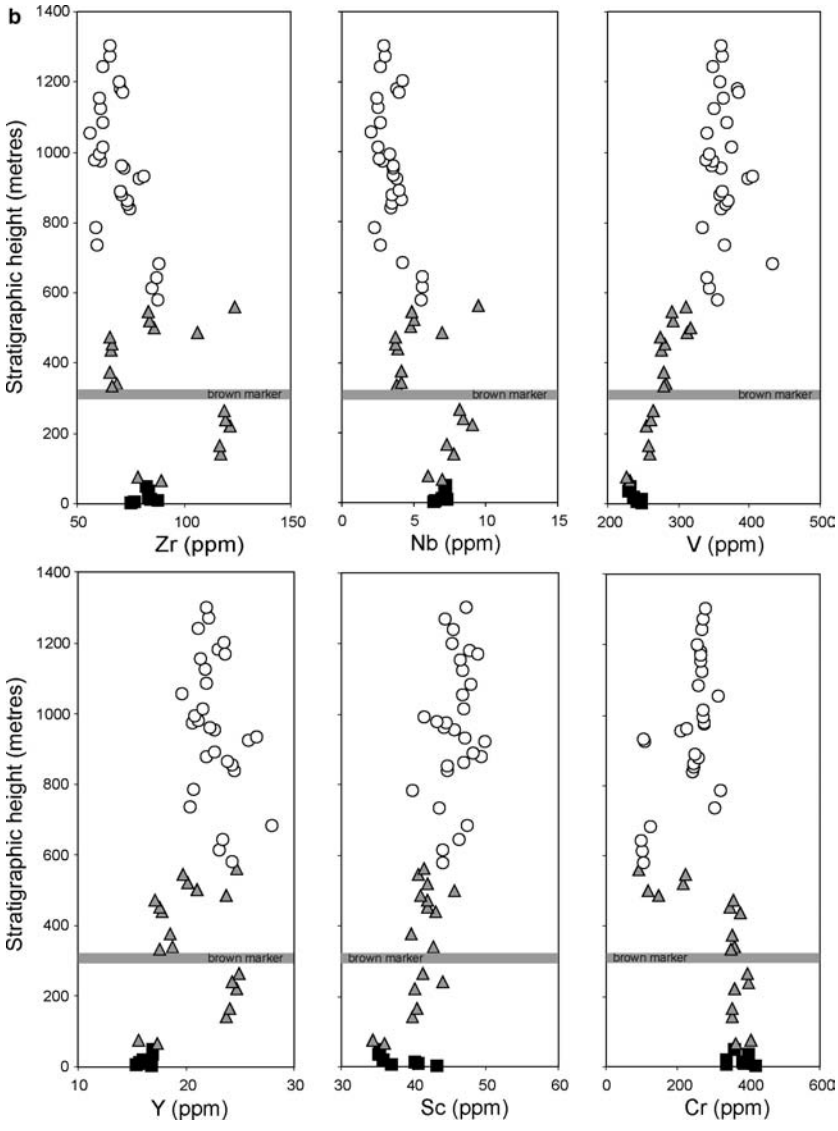
generally mimics the behaviour of Fe_2O_3. The lavas comprise low titanium basalts, with mean TiO_2 contents rising through the Basal, Aphyric and Porphyritic units with values of 0.87, 0.91 and 1.05 wt.%, respectively.

Despite considerable scatter, Al_2O_3 shows generalised decrease through the Basal and Aphyric units up to ~200 m. Above this Al_2O_3 contents vary erratically at ~14 wt.% through the Porphyritic unit. Any original CaO trends appear to have been severely affected by differential leaching during post-eruptive alteration (Kalsbeek and Jepsen 1984). There is a strong negative covariation between CaO and the proportion of volatiles, the least altered lavas (Loi2O_5 content characterises the entire sequence but with a notable fall between 300 and 500 m. An increase at ~500 m is followed by a very generalised decrease through the Porphyritic unit (Fig. 4a).

Plots of Zr, Nb and Y versus stratigraphic height show that basalts immediately above the 'brown marker' (ca. 300 m up) have strikingly reduced contents of Zr, Nb and Y relative to those below (Fig. 4b). P_2O_5 and TiO_2 contents are also reduced in basalt lavas above this stratum (Fig. 4a). V displays fairly steady up-sequence increase from ~220 to ~360 ppm. Sc, after an initial decrease in the Basal unit increases slightly from ~34 to ~46 ppm through the Aphyric and Porphyritic units. Nb and Cr show overall decrease upwards although Cr exhibits a striking fall from ~400 to ~90 ppm at ~500 m but with restoration of values at ~700 m to ~260 ppm.

Fig. 5 presents variation diagrams for some major and trace elements versus Mg number. On these plots $Fe_2O_3^T$, TiO_2, Y and V display consistent increase as Mg number declines. Zn and Sc behave similarly although with distinctly more scattered trends. At similar Mg numbers samples from the Porphyritic unit have markedly higher concentrations of $Fe_2O_3^T$, V and Ni than those of the Aphyric unit. La/Lu values decrease from the Basal unit (>40) through the Aphyric unit to attain a fairly constant value of <20 in the Porphyritic unit

Fig. 4 (Contd.)



(Fig. 6a). There is a positive correlation between Zr/Nb and stratigraphic height with the greatest scatter occurring in the Porphyritic unit (Fig. 6a).

La/Nb ratios are typically in the range 1.5–2.0, with mean values rising from 1.43–1.89 to 1.93 through the three units. The negative correlation between La/Lu and Zr/Nb, progressive through the three units, is shown in Fig. 6b. Chondrite-normalised REE patterns for representative basalts from the three units are shown in Fig. 7. In the Basal unit there is a small fall in normalised LREE from La to Gd and thereafter values are essentially constant through the MREE to HREE. The patterns for the Aphyric unit basalts are broadly similar to these but with higher normalised values for the HREE. Basalts from the Porphyritic unit, however, have distinctly flatter patterns than those of the earlier units, showing very little LREE enrichment. La_N/Lu_N values show progressive decrease through the three units (Basal unit, 4.76–4.68: Aphyric unit, 4.22–2.53: Porphyritic unit, 2.03–1.48).

Incompatible element patterns normalised against depleted mantle for representative basalts and dolerites are shown in Fig. 8. Because of their mobility during hydrothermal alteration the concentrations of Ba, Rb, K and Sr scatter very widely. The rocks typically display small negative anomalies for Nb, P and Ti. Between Nb and Lu there is a sequential decrease in slope (Nb/Lu) shown by the three units, steepest for the Basal unit and notably shallower for the Porphyritic unit.

Compositions of the Midsommersø dolerites

The intrusive suite beneath the Zig-Zag Dal lavas shows wide variation from mafic to salic compositions (Kalsbeek and Jepsen 1983). Relative ages within the intrusive suite are poorly known and, probably because of the combination of fractional crystallisation, contamination and hydrothermal alteration, variation diagrams indicate broad scatter and are not shown. The composi-

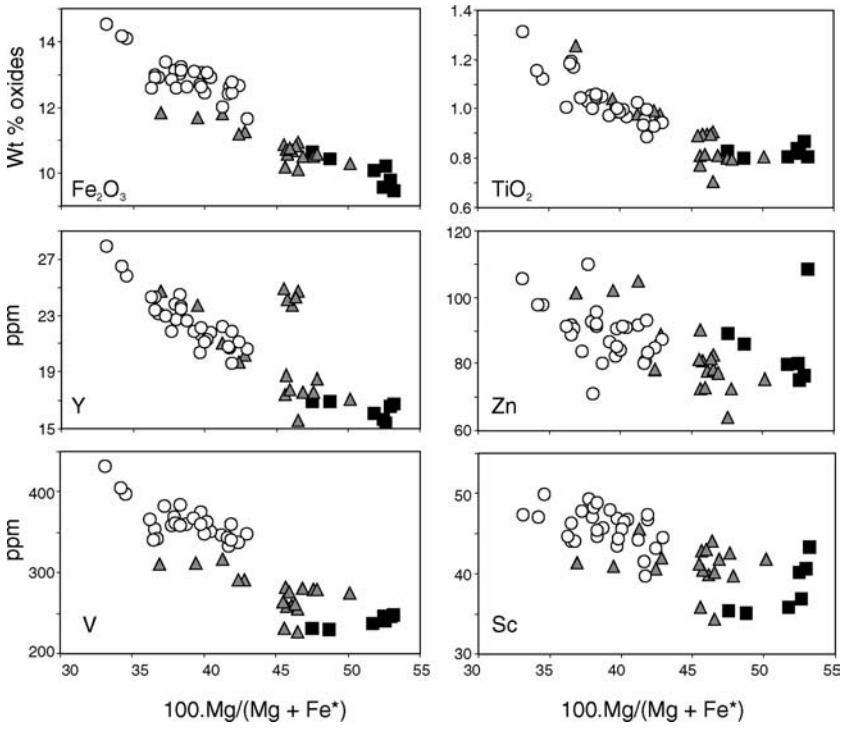


Fig. 5 $Fe_2O_3^T$, TiO_2, Y, Zn, V and Sc versus Mg number. Symbols as in Fig. 3

tional overlap, however, with data from the Zig-Zag Dal lavas is extensive, supporting the proposition that the intrusive and extrusive suites are approximately contemporaneous and cogenetic.

Chondrite-normalised REE and MORB mantle-normalised patterns for three dolerite samples (Figs. 7, 8) are generally comparable to those of the lavas. The La_N/Lu_N range of 4.05–2.89 for the Midsommersø intrusions most closely accords with that of the Aphyric unit basalts. The MORB mantle-normalised incompatible element patterns for Midsommersø samples (Fig. 8) also most closely match those of the Aphyric unit basalts, with relative deficiencies of Nb, P and Ti.

Whole-rock isotopic data

Neodymium isotopes

Ten samples from the Zig-Zag Dal lavas were analysed for Nd isotopes. The samples are moderately to strongly enriched in light REE with ^{147}Sm–^{144}Nd ratios ranging from 0.123 to 0.181 and $^{143}Nd/^{144}Nd$ values between 0.5118 and 0.5125 (Table 3). Calculated at 1382 Ma, they have ε_{Nd} values of +0.7 to −3.9 and $T_{DM}Nd$ model ages (DePaolo 1981) of 2.12–2.41 Ga (Table 3). If, as is supposed, these basalts are coeval with the underlying Midsommersø dolerites, they must have had variable initial isotopic compositions.

Four samples of the Midsommersø sills have ^{147}Sm–^{144}Nd ratios from 0.125 to 0.169 and $^{143}Nd/^{144}Nd$ values from 0.5117 to 0.5124 and the data show wide scatter in the $^{147}Sm/^{144}Nd$ versus $^{143}Nd/^{144}Nd$ diagram (Fig. 9a). Single sample ε_{Nd} values range from −5.2 to + 0.9 at 1382 Ma and indicate substantial variation in the isotopic compositions of the dolerite magmas. The Nd isotope composition of one of the dolerite samples (GGU 197402) is quite similar to that of the less radiogenic (low ε_{Nd}) Zig-Zag Dal lavas, while samples 273251 and 335774 are akin to the more radiogenic (high ε_{Nd}) lavas in this respect. The latter two samples show that a chondritic mantle was involved in the Midsommersø magmatism. The composition of sample GGU 273493 is clearly less radiogenic (ε_{Nd} at 1382 Ma is −5.2) than the three other Midsommersø dolerite samples (Table 3).

For the basalts there is a statistically highly significant negative correlation between the ε_{Nd} values and the SiO_2 contents of the analysed basalts. The same is true for the dolerites; GGU 273493, with 60.6% SiO_2, is the most unradiogenic of the samples studied.

Hafnium isotopes

All ten Zig-Zag Dal basalts have sub-chondritic Lu/Hf and initial (at 1,382 Ma) ε_{Hf} values that vary from +4.4 to −3.9, a range which compares well with the Nd isotope data for these lavas. The four Midsommersø sill samples are likewise characterised by sub-chondritic Lu/Hf and have ε_{Hf} varying from +4.3 to −7.2, again closely matching the corresponding Nd isotope variation. As the correlation between Hf and Nd isotopes for this suite of samples is remarkably good, all observations made in the previous paragraph for Nd systematics also hold true for those of Hf.

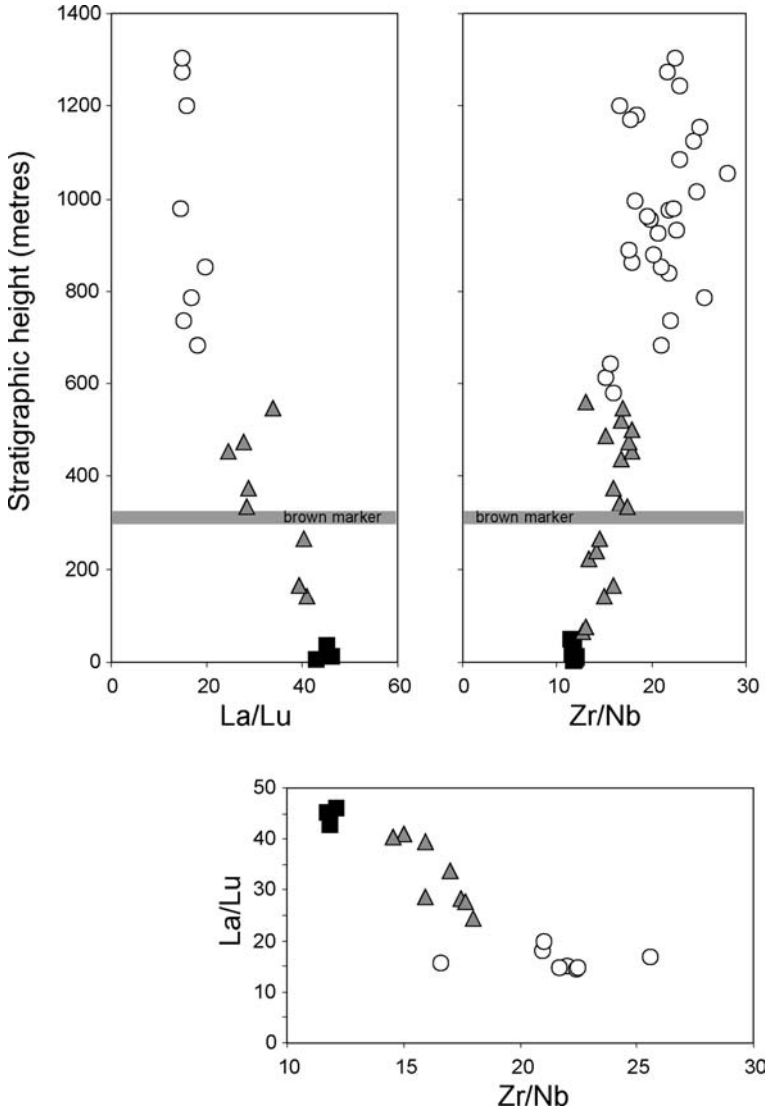


Fig. 6 **a** La/Lu and Zr/Nb versus stratigraphic height. **b** La/Lu versus Zr/Nb. Symbols as for Fig. 4.

Sm–Nd and Lu–Hf errorchrons and initial ratios

When the Nd isotopic data from all fourteen samples are regressed, the result is a very poorly defined (MSWD = 69) trend, the slope of which corresponds to an age of 2,023 ± 310 Ma (Fig. 9a). The ten lava samples yield an errorchron with a better fit (MSWD = 11.3) but a geologically unacceptable age of 2,025 ± 150 Ma. The data show that, in terms of initial Nd isotope composition, the Zig-Zag Dal lavas fall into two sub-groups:

1. Low ε_{Nd} samples (273405–273436, from the Basal and Aphyric units) with ε_{Nd} (at 1,382 Ma) between −3.9 and −2.3.
2. High ε_{Nd} samples (273442–273470, from the Porphyritic unit) with ε_{Nd} (at 1,382 Ma) between −0.7 and +0.7.

Individual regressions of these sub-groups yield ages of 1736 ± 410 Ma (MSWD = 8.5) and 1,817 ± 560 Ma (MSWD = 4.2), respectively (Fig. 9a).

Using the Hf isotope data, the fourteen samples yield a poor trend (MSWD = 159) that conforms to an age of 2,366 ± 510 Ma. In terms of initial ratios at 1,382 Ma, the lava samples also define two sub-groups complying with the twofold grouping that emerged from the Nd isotope compositions (Table 3; Fig 9b). These two sub-groups yield trends that correspond to ages of 2,116 ± 340 Ma (MSWD = 4.5; low ε_{Hf} group) and 2,019 ± 580 Ma (MSWD = 3.6; high ε_{Hf} group), (Fig. 9b).

Nd–Hf correlations

Table 3 shows the initial ε_{Nd} and ε_{Hf} values for all 14 samples calculated at 1,382 Ma. Figure 10 shows these values in an ε_{Nd}–ε_{Hf} variation diagram. The two basalt sub-groups (see above) form two distinct populations:

1. Low ε_{Nd}—low ε_{Hf} (unradiogenic basalts from the Basal and Aphyric units).

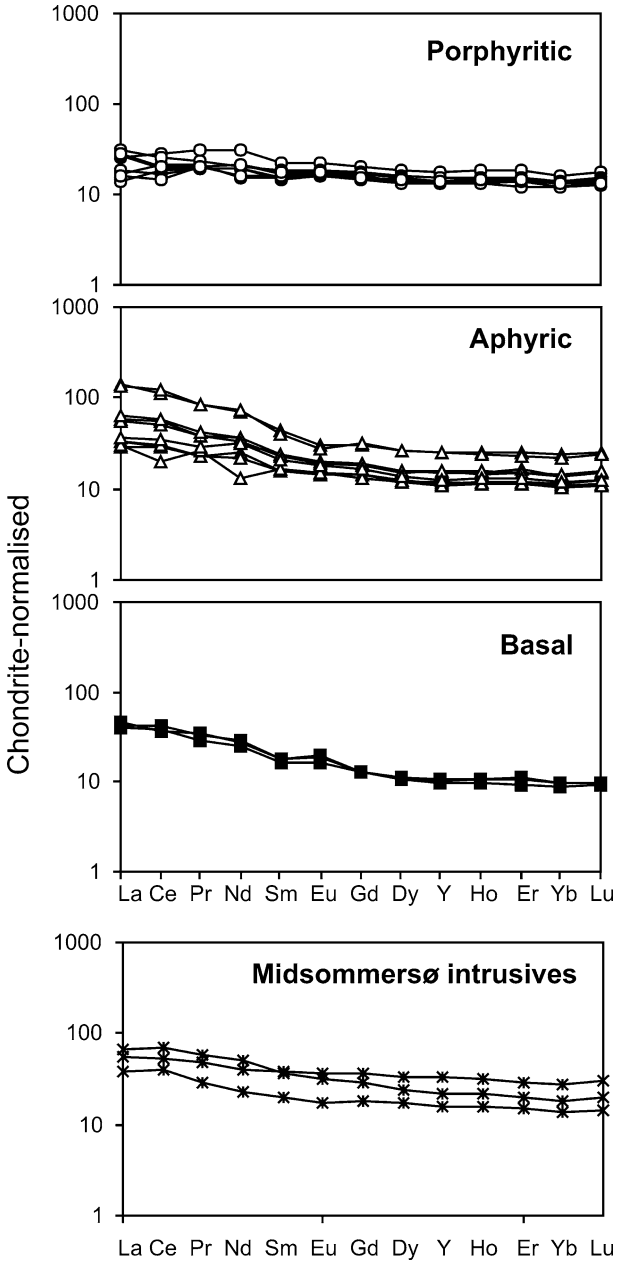


Fig. 7 Chondrite-normalised REE patterns for selected samples from the Midsommersø intrusions and the three units in the lava succession. Normalising factors after Nakamura 1974

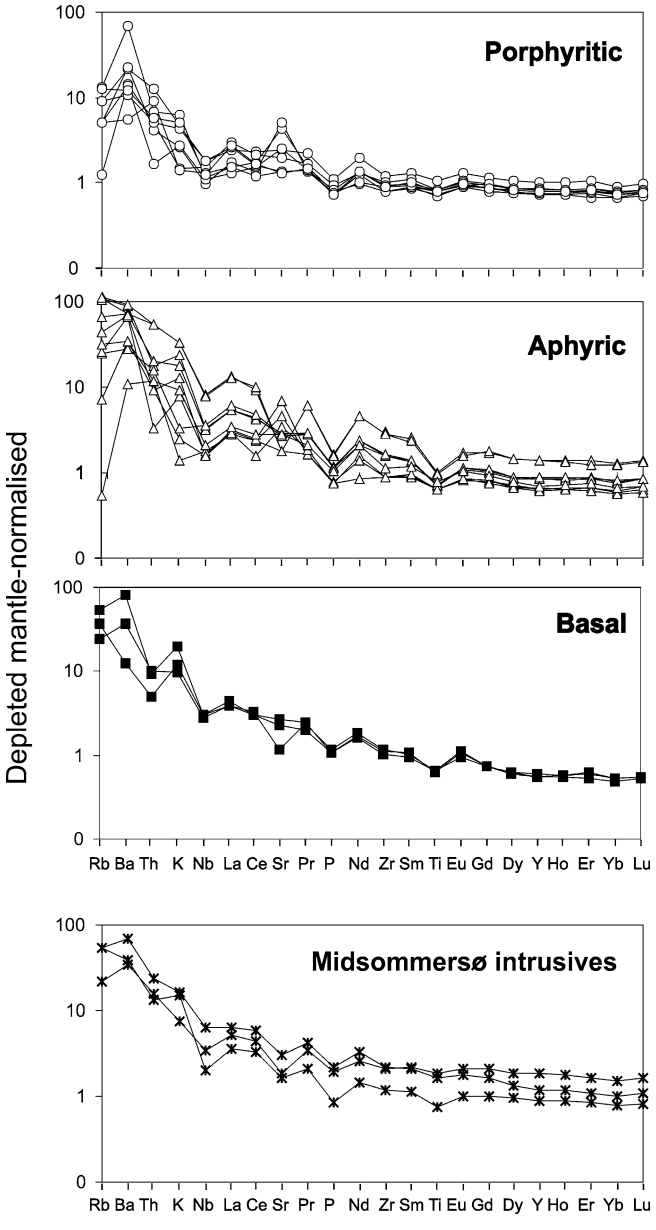


Fig. 8 Depleted mantle-normalised incompatible element plots (normalising factors after Sun and McDonough 1989)

2. High ε_{Nd}—high ε_{Hf} (radiogenic basalts from the Porphyritic unit).

Two of the dolerites (273251, 335774) conform to the high ε_{Nd}—high ε_{Hf} population, one (197402) to the low ε_{Nd}—low ε_{Hf} population and one (273493) is distinctly more unradiogenic than all of the other samples. GGU 273493 is the SiO_2-rich sample mentioned above (Table 1) and the low epsilon value is probably due to crustal contamination. The variation in the ε_{Nd} (1,382 Ma) values of three of the radiogenic basalt samples and the two radiogenic dolerites (+0.3 to +0.9) is less than the maximum determinative error of ±0.4 ε-units.

Discussion

High MgO (>10 wt.%) lavas occur in numerous continental flood basalt (CFB) successions, commonly at the base and the restriction of the higher MgO values to the Basal unit lavas in the Zig-Zag Dal sequence conforms to this pattern. While relatively primitive magmas can reach surface levels in the initial, high energy stages of the eruptive cycle, later magma batches are increasingly liable to retention within lower crustal magma

Table 3 Nd and Hf isotope data for the Zig-Zag Dal basalts and Midsommersø dolerites, NE Greenland

Sample	Sm [a] (ppm)	Nd [a] (ppm)	^{147}Sm [b]/ ^{144}Nd	^{143}Nd [c]/ ^{144}Nd	ε_{Nd} [d] (T)	T_{DM} [d] (Ma)	Lu [a] (ppm)	Hf [a] (ppm)	^{176}Lu [b]/^{177}Hf	^{176}Hf [e]/^{177}Hf	ε_{Hf} [f] (T)
Zig-Zag Dal											
Basal unit											
273405	2.58	11.78	0.1321	0.511899 ± 9	−3.0	2131	0.5749	5.645	0.01445	0.282154 ± 5	−3.9
273410	3.28	14.89	0.1332	0.511872 ± 12	−3.7	2212	0.2395	2.026	0.01678	0.282263 ± 6	−2.3
273419	3.74	16.61	0.1361	0.511889 ± 10	−3.9	2264	0.3409	2.945	0.01643	0.282246 ± 5	−2.6
Aphyric unit											
273427	6.46	31.79	0.1228	0.511775 ± 9	−3.8	2120	0.2317	1.912	0.01720	0.282290 ± 6	−1.8
273434	2.42	9.55	0.1533	0.512125 ± 8	−2.3	2316	0.2545	1.748	0.02066	0.282416 ± 5	−0.6
273436	3.65	16.00	0.1377	0.511937 ± 8	−3.2	2215	0.3333	2.758	0.01715	0.282292 ± 6	−1.6
Porphyritic unit											
273442	3.44	13.28	0.1564	0.512238 ± 9	−0.7	2136	0.3332	2.283	0.02072	0.282494 ± 6	+2.1
273447	2.55	8.65	0.1780	0.512485 ± 13	+0.3	2393	0.2967	1.636	0.02574	0.282693 ± 7	+4.4
273451	2.85	10.30	0.1675	0.512341 ± 11	+0.3	2319	0.3363	1.953	0.02443	0.282630 ± 5	+3.4
273470	2.77	9.24	0.1813	0.512534 ± 10	+0.7	2409	0.3351	1.910	0.02491	0.282670 ± 5	+4.3
Midsommersø sills											
197402	3.15	12.41	0.1534	0.512104 ± 22	−2.8	2375	0.3550	2.400	0.02099	0.282419 ± 5	−0.8
273251	6.09	21.81	0.1687	0.512411 ± 10	+0.5	2137	0.7124	4.444	0.02275	0.282573 ± 6	+3.0
273493	3.06	14.78	0.1250	0.511724 ± 10	−5.2	2262	0.2922	2.825	0.01468	0.282069 ± 6	−7.2
335774	5.69	24.89	0.1382	0.512154 ± 9	+0.9	1786	0.4674	3.879	0.01710	0.282408 ± 6	+2.5
335774, duplicate							0.4691	3.883	0.01715	0.282458 ± 6	+4.3

[a]Sm, Nd, Lu, and Hf concentrations by ID/MS

[b]2σ error on the concentration ratios is $< 0.5\%$

[c]$^{143}Nd/^{144}Nd$ normalised to $^{146}Nd/^{144}Nd = 0.7219$; uncertainties reported are $2\sigma_m$ in the last significant digits. $^{143}Nd/^{144}Nd$ of La Jolla Nd standard = 0.511850 ± 0.000015 (2σ, $n = 10$) (i.e., external reproducibility = 25 ppm)

[d]Initial ε_{Nd} values at 1382 Ma, calculated using chondritic values of $^{143}Nd/^{144}Nd = 0.51264$ and $^{147}Sm/^{144}Nd = 0.1966$; T_{DM} is depleted mantle model age (DePaolo 1981)

[e]$^{176}Hf/^{177}Hf$ normalised to $^{179}Hf/^{177}Hf = 0.7325$; uncertainties reported are $2\sigma_m$ in the last significant digits. $^{176}Hf/^{177}Hf$ of JMC-475 Hf standard = 0.282160 ± 0.000010 (2σ) (i.e., external reproducibility = 35 ppm). Hf standard run alternately with samples

[f]Initial ε_{Hf} values at 1382 Ma, calculated using $\lambda = 1.93 \times 10^{-11}$ years^{-1} and chondritic values of $^{176}Hf/^{177}Hf = 0.282772$ and $^{176}Lu/^{177}Hf = 0.0332$

chambers to undergo more prolonged fractionation prior to ascent and eruption (*cf.* Cox 1980; Gibson et al. 2000). The majority of the Zig-Zag basalts, like most CFBs, have between 5 and 8 wt.% MgO. As has been argued (e.g. Farnetarni et al. 1996) such melts would not have been in equilibrium with mantle peridotites and are most probably residuals after crystal fractionation. The latter is likely to have mainly occurred in the vicinity of the crust–mantle boundary with concomitant production of underplating cumulates.

Previous work by Kalsbeek and Jepsen (1984) concluded that the hydrothermal metamorphism (incipient spilitisation) of the basalts seriously limits the use of oxide–oxide variation plots. Accordingly their conclusions regarding the effects of fractional crystallisation on the composition of the lavas were largely based on Pearce-type variation diagrams (Pearce 1968) where major elements were used in conjunction with the incompatible elements P and Zr. Chemical mixing calculations (Wright and Doherty 1970) were used to check the conclusions reached from the Pearce diagrams. They deduced that much of the compositional variations within each of the three units were explicable in terms of variable degrees of low-pressure fractional crystallisation of plagioclase, augite and olivine. However, they noted that the compositional breaks coincident with the boundaries defining the three units, and also between the basalts overlying and underlying the 'brown marker' in the Aphyric unit, were not explicable in terms of fractional crystallisation. These changes they ascribed to heterogeneity of the contributing source regions. Thus, Kalsbeek and Jepsen (1984) concluded from the generalised upward decrease in P_2O_5 and Zr that the magmas had been derived from increasingly depleted mantle sources. The new data largely confirm these conclusions and also introduce some refinements.

Although much of the scatter in Mg numbers can be ascribed to fractional crystallisation, the overall fall in Mg number with stratigraphic height (Fig. 4a) cannot be related to fractional crystallisation of olivine since the behaviour of nickel is not sympathetic to that of magnesium, but tends to increase upwards through the sequence.

Although the passage from Basal to Aphyric unit corresponds to a marked change in chemistry the next most significant change in the basalt compositions is at around 300 m up from the base of the succession (i.e. the level of the 'brown marker'). The distinctly more evolved character of the 'brown marker' tholeiitic andesite suggests that there was a significant decrease in magma productivity allowing time for maturation and fractional crystallisation at depth before ascent of relatively primitive basalt magmas resumed. The fall in Zr, Nb, Y, P and Ti in the post-'brown marker' basalts relative to those preceding it, may relate to an abrupt increase in melt fraction. The change in magma chemistry seen at ca. 500 m up-section involving enhancement in $Fe_2O_3^T$, TiO_2 and (to a lesser extent) V content concomitant with

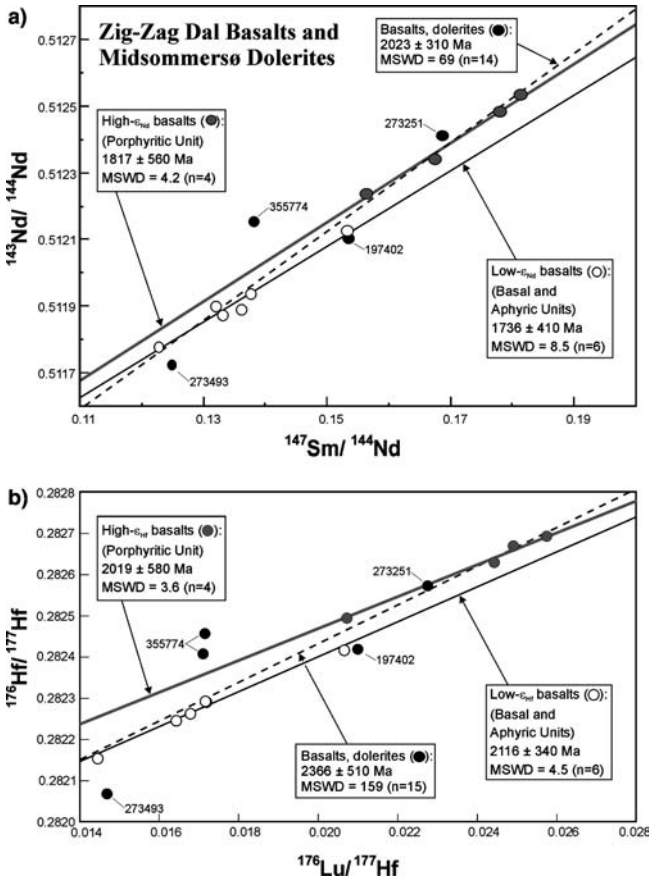


Fig. 9 Sm-Nd (**a**) and Lu–Hf (**b**) isochron diagrams showing the composition of ten Zig-Zag Dal basalts and four Midsommersø dolerites analysed for Nd and Hf isotopes. Least-square fitted trends of the entire data set (*dashed line*) and the two sub-populations (*solid lines*) are shown with the corresponding ages and pertinent statistical parameters. Dolerite samples are identified (see Table 3)

a fall in MgO wt.% and in Mg number, may be the result of more advanced fractional crystallisation. These lavas also have enhanced SiO_2 contents (Fig. 4a) and crystal fractionation may have been accompanied by a minor degree of crustal contamination.

Iron-enrichment in tholeiitic basalt evolution is usually explicable as a consequence of plagioclase fractionation. However, while it is accepted that the magmas that gave rise to the Porphyritic unit, like those of the Aphyric (and possibly Basal) units, had experienced some plagioclase fractionation (Kalsbeek and Jepsen 1984), this did not produce any significant Eu anomaly. Consequently it is unlikely to have been the cause of the overall Fe enrichment observed throughout the sequence (Fig 4a). The steady increase of Fe, Ti and V also implies that magnetite fractionation was not involved.

The increase in Zr/Nb up section suggests that either the extent of melting in the mantle source increased with time because Nb has a lower bulk distribution coefficient than Zr in relation to typical mantle mineralogy and/or that there was an increased contribution of melts low in Nb. La/Lu values decrease with stratigraphic height (Fig. 5a) and, because the HREE are more compatible than the LREE, this provides additional evidence that melt fractions increased with time. However, it could also be explained in terms of decreasing contamination by components derived from lithospheric mantle or crustal sources that were relatively rich in LREE and Zr. While the Nd and Hf isotope data tend to support this argument, it does not necessarily contradict the model of increasing source melting. The two processes may have operated in conjunction.

Fig. 11 illustrates Ce/Y versus Zr/Nb for the basalt lavas from the type Zig-Zag Dal section and, like Fig 5b, it displays the consistent variation through the three units. The data are plotted together with non-modal equilibrium melting curves for primitive and depleted mantle. The Basal unit data fall closest to the primitive mantle curve (garnet lherzolite facies) and mid-way between the curves for depleted garnet lherzolite and depleted spinel lherzolite. The bulk of the Aphyric unit data also lie roughly midway between the curves for depleted garnet lherzolite and depleted spinel lherzolite but offset in the direction indicative of greater melt fractions. By contrast, the Porphyritic unit data points closely approximate to the depleted spinel lherzolite curve. The plot may be used to support the contention that not only was the melt fraction increasing with time but that melting was occurring at progressively shallower levels, becoming dominated by melts in the spinel lherzolite stability field by the time of Porphyritic unit production. Use of the Ce/Y versus Zr/Nb figure has the limitation, however, of implying that all the melting was sub-lithospheric and that any lithospheric contribution was negligible.

Figure 12 shows a comparable Ce/Y versus Zr/Nb diagram for the Midsommersø dolerites. Although the scatter is large, the sill data show broad overlap with those of the overlying lavas. The discrepancy between the two data sets is not sufficient to show that the magmas for lavas and the intrusions were not cognate. As noted in the discussion of age data, the possibility remains that the dolerite sills reflect more than one event.

The Zig-Zag Dal formation in relation to other continental flood basalt successions

Whilst there is no independent evidence to support the suggestion that a plume was responsible for the Zig-Zag Dal—Midsommersø magmatism, the data are compatible with this hypothesis. Anand et al. (2003), presented an argument for the (ca. 2000 Ma) Cuddapah Basin volcanic series in NE India, claiming that it was not necessary to attribute the volcanism to a thermal plume since at that relatively early stage in the Proterozoic there was still enough thermal energy for the requisite melting to have been due to lithospheric extension alone. However, the significantly younger Zig-Zag Dal event produced a succession with so many features in common

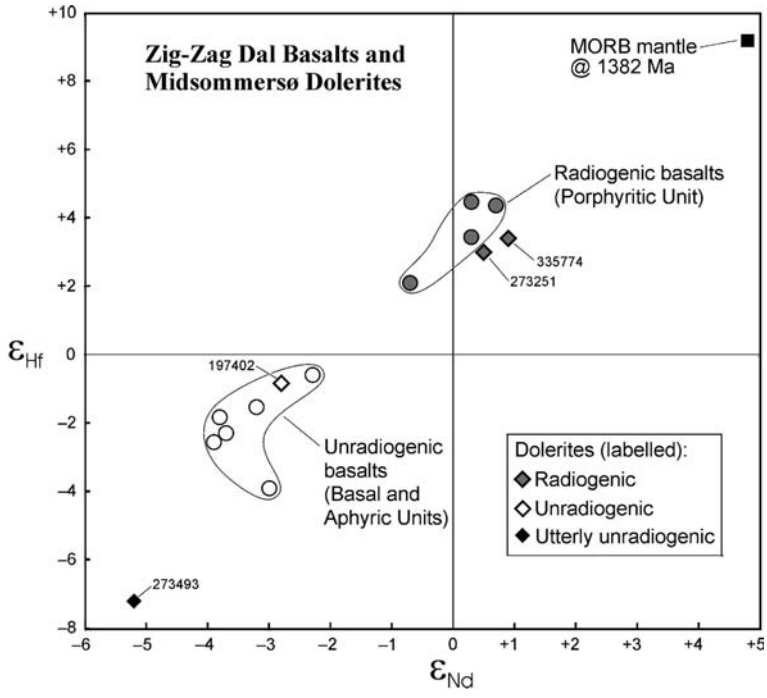


Fig. 10 ε_{Hf} versus ε_{Nd} diagram showing the initial (at 1,382 Ma) Hf and Nd isotope compositions of Zig-Zag Dal basalts and Midsommersø dolerites. MORB mantle at 1382 Ma was calculated using $^{176}Lu/^{177}Hf = 0.0332$, $^{147}Sm/^{144}Nd = 0.1296$, $^{176}Hf/^{177}Hf = 0.282772$ and $^{143}Nd/^{144}Nd = 0.512634$

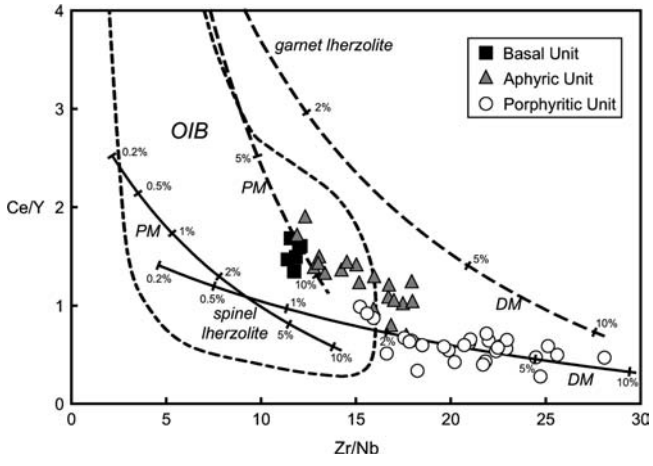


Fig. 11 Ce/Y versus Zr/Nb plot for basaltic lavas in the type section, Zig-Zag Dal. Non-modal equilibrium melting curves (Shaw 1979) are for primitive mantle (PM, composition from McDonough and Sun 1995) and broken curves are for depleted mantle (DM, composition calculated from the average N-MORB composition given by Sun and McDonough 1989). Mantle and melting modes are from Johnson (1998). Distribution coefficients for garnet and clinopyroxene are from Johnson (1998): olivine and orthopyroxene coefficients from Bedini and Bodinier (1999); spinel coefficients from Stracke et al. (2003). The ocean-island basalt (*OIB*) field is based on unpublished data (J. G. Fitton and D. James). Symbols as in Fig. 3

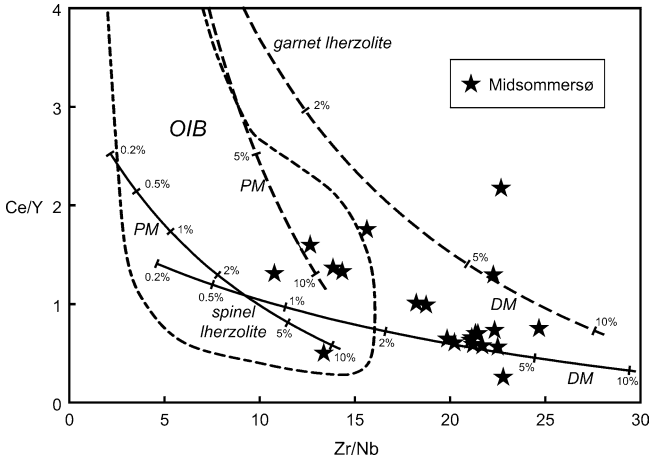


Fig. 12 Ce/Y versus Zr/Nb plot for Midsommersø dolerites. Otherwise, as for Fig. 11

with Phanerozoic CFB analogues that involvement of a mantle plume remains the preferred hypothesis.

Although some authors (e.g. Saunders et al. 1992) considered that most CFBs record a trace element and isotopic contribution from the lithosphere through which they erupted, this is disputed by Farnetarni and Richards (1994). The latter authors concluded that some 90% of plume melting proceeds at sub-lithospheric depths (> 100 km) since there would be insufficient time for conductive heating of the lithosphere to temperatures above its solidus before the plume-head had spread and cooled.

Initial work on the Zig-Zag Dal lavas led Kalsbeek and Jepsen (1984) to conclude that different mantle sources must have been involved. Normalised incompatible element patterns (Fig. 7) are grossly different from those of oceanic regions assumed to arise from asthenospheric mantle (*cf.* Sun and McDonough 1989).

Involvement of the lithospheric mantle has commonly been invoked as a factor in the petrogenesis of CFBs, as for example the Deccan Traps (Melluso et al. 1995). Arndt and Christensen (1992) proposed a model in order to try to account for the widespread negative Nb

anomalies in CFBs (such as are exhibited by the basalts under discussion) which postulates that sub-lithospheric melts acquire some of their geochemical characteristics through partial reaction and chemical exchange with lithospheric mantle wall rocks. A study of Swedish Proterozoic basic intrusions (Patchett et al. 1994) also concluded that asthenospherically derived melts acquired their variable ε_{Nd} values and negative Nb anomalies from reaction with the lithosphere. Features noted by Patchett et al. (1994) that were inexplicable by crustal contamination or crystallisation processes included the variation in incompatible element patterns and ε_{Nd} from -0.5 to $+3.5$. These authors (*op. cit*) suggested that the magmas came either from lithospheric mantle enriched by incompatible elements derived from a subducting slab or that they had an asthenospheric origin and acquired their incompatible elements (but relatively little Nb) through interaction with lithospheric mantle.

For the Zig-Zag Dal magmatism we suggest that melting of asthenospheric garnet lherzolite-facies peridotites commenced at depths of 120–70 km (Farnetarni et al. 1996). Progressive thinning of the lithosphere through extensional tectonics, possibly accompanied by mechanical or thermal erosion, then permitted the hypothesised plume head to ascend and decompress further so that melting continued into the spinel lherzolite stability field at levels as shallow as 35–50 km. This model is comparable to that presented by Kerr (1994) with respect to the North Atlantic Tertiary Province.

In general, initial Nd and Hf isotope compositions in the Zig-Zag Dal and Midsommersø rocks correlate positively (Table 3, Fig. 10), reflecting the congruent behaviour of these two isotopic systems during mantle melting. The two distinct ε_{Nd}–ε_{Hf} populations are compatible with the concept of two distinct sources for the melts. The earlier (Basal and Aphyric) basalts (and dolerite sample 197402) with low ε_{Nd}–low ε_{Hf} could either have been derived from an enriched source or have been more contaminated by lithospheric interaction. The negative covariation of ε_{Nd} and ε_{Hf} values with SiO_2 contents suggests that a minor degree of crustal contamination may also have taken place. In contrast, the younger (Porphyritic) basalts and dolerites (273251 and 335774), characterised by higher ε_{Nd}–higher ε_{Hf}, may have come from a slightly depleted or less contaminated source. An overall up-sequence decrease in $^{87}Sr/^{86}Sr$ ratio was reported for the lavas by Kalsbeek and Jepsen (1984) lending some support to the thesis that contamination lessened as the activity proceeded.

The marked increase in Fe in the Porphyritic unit, relative to the Basal and Aphyric units, cannot be ascribed to fractional crystallisation of a Fe-poor mineral assemblage because it is not accompanied by a change in Mg numbers (Fig. 5a). Rather, it may reflect involvement of a more Fe-rich source from that stage on. The situation is not dissimilar to that described for the Lebombo (Zimbabwe/Mozambique) basalts of the Karoo which were subdivided into three groups regarded as products of the same plume (Sweeney et al. 1994). Genesis of the two earlier basaltic groups involved extensive lithospheric interaction whilst the later, high-Fe basalts, were inferred to more closely represent the plume source. Similarly in the Zig-Zag Dal, the least contaminated basalts (on the basis of their ε_{Nd} values) are the youngest. As in the Lebombo case, transition up through the Basal, Aphyric and Porphyritic units could reflect decreasing degrees of lithospheric contamination with the youngest (porphyritic) lavas being products of essentially uncontaminated plume-source magmas.

A plume-source mantle with abnormally high Fe/Mg was proposed by Scarrow and Cox (1995) as a result of investigation of primitive basalts from Skye (Hebrides) in which a strong negative correlation between Si and Fe was observed. The antipathetic relationship of Si and Fe (Fig. 3b) in the Zig-Zag Dal lavas indicates some affinity to the Skye situation. Turner and Hawkesworth (1995) also concluded that melts produced by plumes beneath continental lithosphere have relatively low Si and high Fe whereas partial melts with low Ti and Fe (among others) come from peridotites previously depleted by melt extraction. Whereas dry depleted peridotites (such as may typify the lithospheric mantle) have a high-temperature solidus, CFBs may originate through melting of less refractory, variably-depleted but hydrated lithospheric mantle sources (*cf*. Saunders et al. 1992). Isotopic data indicate that such sources were often old and relatively incompatible element enriched and that probably $H_2O \pm CO_2$ was added at the same time as the incompatible elements.

A further case providing evidence for an iron-rich mantle source is that of the rising plume head of the Tristan mantle plume in which Fe-rich 'peridotite streaks' were hypothesised (Thompson et al. 2001). Accordingly it is possible that a rising and expanding melt column developed beneath the NE Greenland crust in the mid-Proterozoic. Early melt fractions were contaminated by incompatible element-enriched fractions derived from the lithospheric mantle or crustal sources but subsequently, as melting progressed, a stage was reached when the contribution from an iron-rich plume component increased significantly giving rise to the magmas from which the Porphyritic unit basalts were produced.

Mesoproterozoic analogues for the Zig-Zag Dal/Midsommersø association

A number of authors have pointed out the similarities between Phanerozoic CFB successions and those of the Mesoproterozoic from ca. 1,400–1,000 Ma (e.g. Hutchison et al. 1990). Possible Mesoproterozoic analogues of the Zig-Zag Dal basalts include the Coppermine River basalts of northern Canada (Baragar 1969), erupted over a short time-scale at ca. 1270 Ma. The 2000–3500 m thick Coppermine succession overlies clastic sediments grading from fluvial to shallow marine at the onset of volcanism (Dostal et al. 1983). Although some crustal contamination is indicated, the mantle component is

inferred to include at least two end-members (Dupuy et al. 1992). Compositional changes up through the section include increase in Fe and Ti and decrease in Mg, Cr, Ni and K. The basalts have negative Nb anomalies similar to those observed in many CFBs and which are characteristic of average continental crust (Thompson et al. 1983). Most of the Coppermine River basalts have positive ε_{Nd}, implying derivation from a source depleted in LREE. Dupuy et al. (1992) invoked mixing of two compositionally distinct components, crust and mantle, and suggested that the lithosphere may have been involved in the genesis of the basalts. Their model proposed that the mantle component was ''a well-stirred mélange'' comprising (a) material from a mantle plume and (b) material from the base of the lithosphere. All the magmas were subsequently affected by crustal contamination. Griselin et al. (1996) concluded that the earliest Coppermine River basalts were produced by melting in the garnet stability field at depths >90 km, probably within a sub-lithospheric mantle plume but that the youngest lavas were generated by higher percentage melting in the absence of garnet at shallower depths.

The Zig-Zag Dal basalts erupted significantly earlier than the great basaltic out-pourings of the Midcontinent Rift of North America, dated between 1109 and 1086 Ma (Davis and Paces 1990). The Midcontinent Rift CFBs may have originated from melting of enriched and extensionally thinned lithosphere as a consequence of plume activity in the deeper mantle (Nicholson and Shirey 1990; Hutchison et al. 1990).

Conclusions

The Zig-Zag Dal basalts and the underlying Midsommersø sheet swarm are typical continental flood basalt products. The large-scale eruption of tholeiitic magmas at 1382 Ma is likely to have accompanied lithospheric extension triggered by a rising mantle plume. Rising contents of HREE, Y and Sc through the lava succession imply progressive consumption of residual garnet into the melt. The progressive upward increase in Zr/Nb and concomitant fall in LREE/HREE (Fig. 5a) marked an increasing degree of melting in the mantle sources and a shallowing of the melt column, ultimately yielding the magmas parental to the Porphyritic unit.

Kalsbeek and Jepsen (1984) noted variations in Zig-Zag Dal lava compositions that appeared inexplicable by fractional crystallisation and which could denote mantle source heterogeneity. The new data confirm and amplify this conclusion. The compositional change from the Aphyric unit to the Porphyritic unit is taken to denote a change in the chemistry of the mantle source(s) to one that was richer in Fe and associated siderophile elements.

The isotopic distinction between the Basal and Aphyric units on the one hand and the Porphyritic unit on the other is significant. The low ε_{Nd}–low ε_{Hf} characteristics of the earlier lavas (Basal and Aphyric units) may have been acquired by contamination with components derived from isotopically enriched lithospheric mantle or crustal sources whereas the high ε_{Nd}–high ε_{Hf} signatures of the younger lavas stem from a mildly depleted (plume-head) source as lithospheric attenuation increased. Thus we attribute these features to decreasing lithospheric contamination with time, with the magmas parental to the Porphyritic unit representing essentially uncontaminated plume source.

Negative anomalies for Nb in the MORB-normalised incompatible element patterns suggest either influence of lithospheric mantle modified by earlier subduction processes (Fitton 1995) or contamination by (relatively) Nb-deficient crustal anatectic melts (Thompson et al. 1983). The high abundances of the most incompatible elements in both the extrusive and intrusive rocks, the negative Nb anomalies, relative to depleted mantle (Fig. 8) as well as the decrease in ε_{Nd} values and the concomitant increase in SiO_2 may all be results of crustal contamination. However, since the ε_{Nd} values of the local crust are very much more negative (down to -15 at 1,382 Ma) than those of the basalts (Kalsbeek et al. 1993), minor proportions of crustally derived components would suffice to impose these features.

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